UIDAI Hackathon 2026 – Submission Report

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1. Problem Statement and Approach

The Aadhaar system is one of the most important identification systems in India. It is used not only for new enrolments, but also for updating personal and biometric information over time. The objective of this project is to understand:

- How Aadhaar enrolments and updates are happening over time
- How activity differs across age groups
- How Aadhaar services are distributed across states and districts

To do this, official UIDAI data was cleaned, organised, and analysed using Power BI. The final output is an interactive dashboard with clear visuals that help explain trends and patterns in Aadhaar enrolments and updates.

2. Datasets Used

This project uses **official UIDAI Aadhaar data** downloaded through API folders. Three separate datasets were used.

Aadhaar Enrolment Data

This dataset contains information about new Aadhaar enrolments.
Main columns used:

- Date, State, District, Pin Code and Enrolments by age group (0–5, 5–17, 18+)

Aadhaar Demographic Update Data

This dataset includes updates related to personal details. Main columns used:

- Date, State, District, Pin Code and Demographic updates for age groups 5–17 and 17+

Aadhaar Biometric Update Data

This dataset contains biometric update records. Main columns used:

- Date, State, District, Pin Code and Biometric updates for age groups 5–17 and 17+

All three datasets follow a similar structure, which made comparison and combined analysis easier.

3. Methodology

3.1 Data Loading

Each dataset was loaded into Power BI using a **folder connection**. This allowed multiple API files to be combined automatically. Hidden or system files were removed to avoid errors.

3.2 Data Cleaning

Some data quality issues were observed and corrected:

- Correct data types were applied to dates and numbers.
- State names had spelling mistakes, extra spaces, and old names.
- State names were standardized to ensure accurate state-wise analysis. Examples:
    - "Westbengal", "West Bengal" → "West Bengal"
    - "Orissa" → "Odisha"
    - "Tamilnadu" → "Tamil Nadu"

This step was very important because incorrect state names would have shown duplicate states in maps and charts.

3.3 Age Group Transformation

In the raw data, age groups were stored as separate columns. To make analysis easier:

- These columns were **unpivoted** into a single "AgeGroup" column
- A "Citizens" column was created to store the values. This made it easier to:
    - Compare age groups
    - Use the same visuals across enrolments and updates

3.4 Date Table for Time Analysis

A separate **Date table** was created to support proper time-based analysis such as:

- Monthly trends, Year-to-date (YTD) calculations, MoM Analysis

The Date table was created using the minimum and maximum dates available in the enrolment data and includes:

- Year, Month, Month number, Quarter, Month-year format, Sorting column for correct month order

It was observed that **August does not appear in the analysis**, as there are **no records available for that month in the UIDAI data**. This reflects a data gap in the source and not an error in the analysis.

4. Data Analysis and Visualisation

4.1 Key Metrics

The dashboard shows the overall scale of Aadhaar activity:

- **Total Enrolments:** 5,435,702
- **Total Demographic Updates:** 49,295,187
- **Total Biometric Updates:** 69,763,095

This shows that update activities are much higher than new enrolments, indicating continuous maintenance of Aadhaar records.

4.2 Age Group Insights

Enrolments

- Most enrolments are from the **0–5 age group**, showing strong early registration.
- Lower enrolments in the 18+ group suggest most adults already have Aadhaar.

Demographic Updates

- Mostly from the **17+ age group**, likely due to address or personal detail changes.

Biometric Updates

- High activity in both **5–17** and **17+** groups, showing regular biometric updates.

4.3 Monthly and YTD Trends

Line charts were used to show:

- Monthly activity
- Year-to-date growth

Key observations:

- A steady increase across months
- Clear cumulative growth through the year
- Activity varies by month, showing operational fluctuations

YTD analysis helps understand overall progress instead of focusing only on individual months.

4.4 State and District Analysis

Map visuals were used to show geographic patterns.

Key observations:

- States like **Uttar Pradesh, Maharashtra, Bihar, and Madhya Pradesh** show high Aadhaar activity.
- Higher activity is linked to population size and administrative scale.
- District-level coverage varies across states and update types.

Daily average values were included to allow fair comparison between states.

5. Conclusion

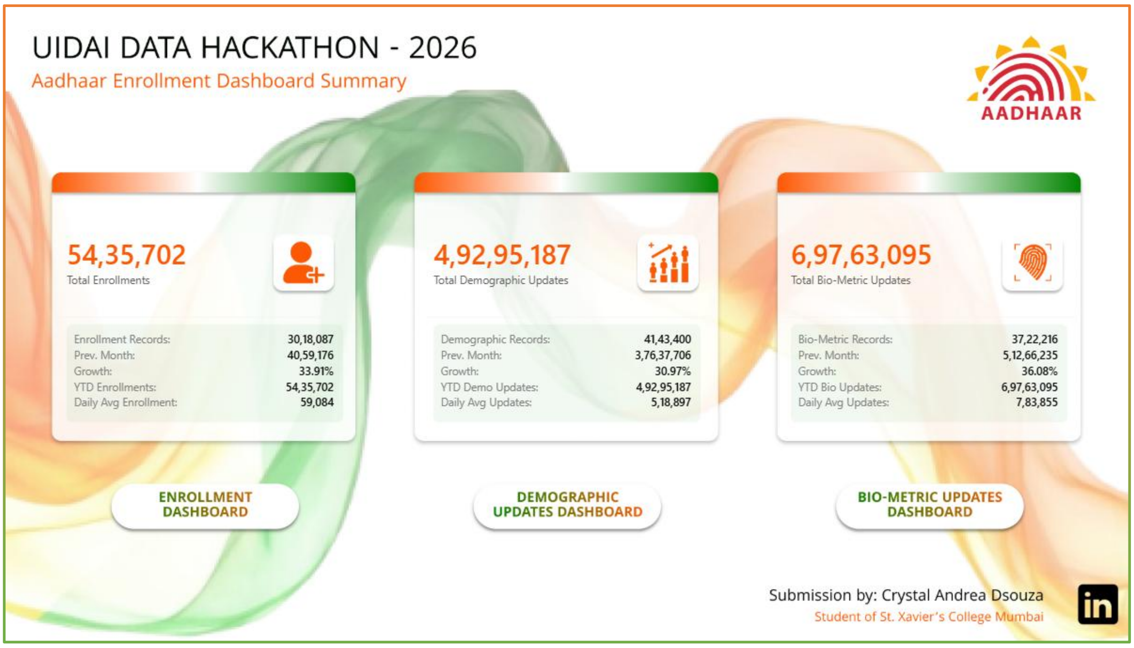
This project shows how raw Aadhaar data can be converted into meaningful insights using proper data cleaning and analysis by:

- Standardizing data
- Creating a structured Date table
- Analysing trends by age, time, and geography

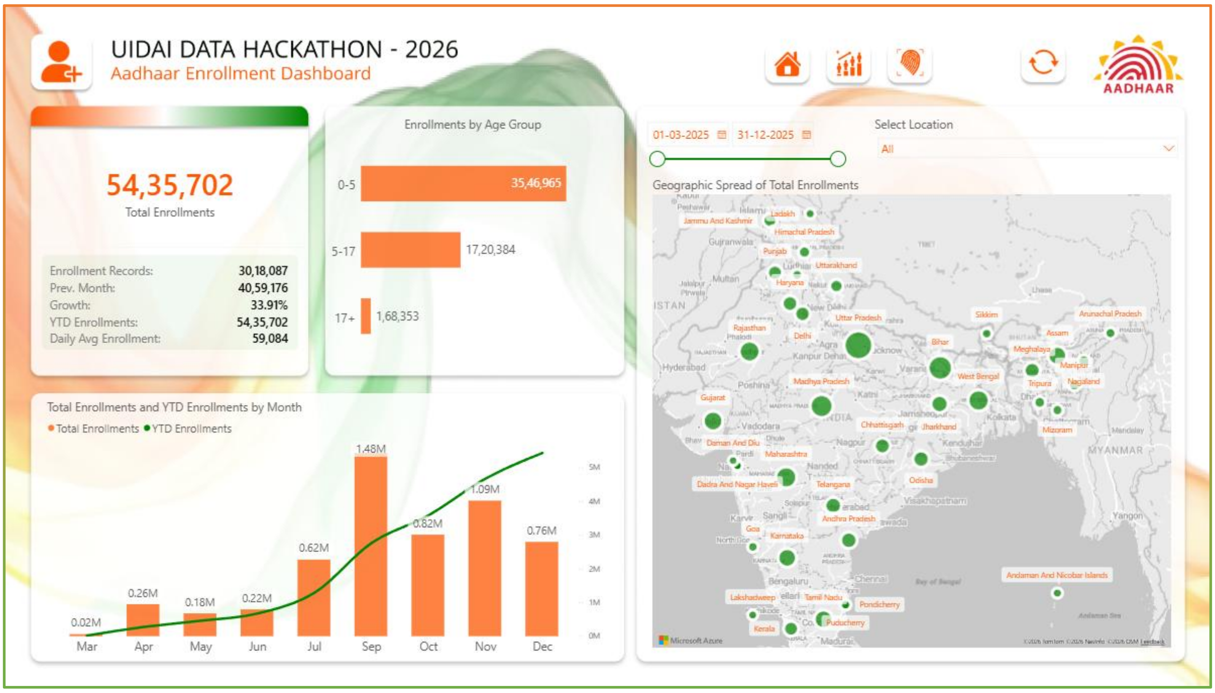
The dashboard provides a clear and easy-to-understand view of Aadhaar enrolments and updates across India. This analysis can help understand service demand and operational patterns within the Aadhaar system.

6. Annexures: Dashboard & Data Model Images

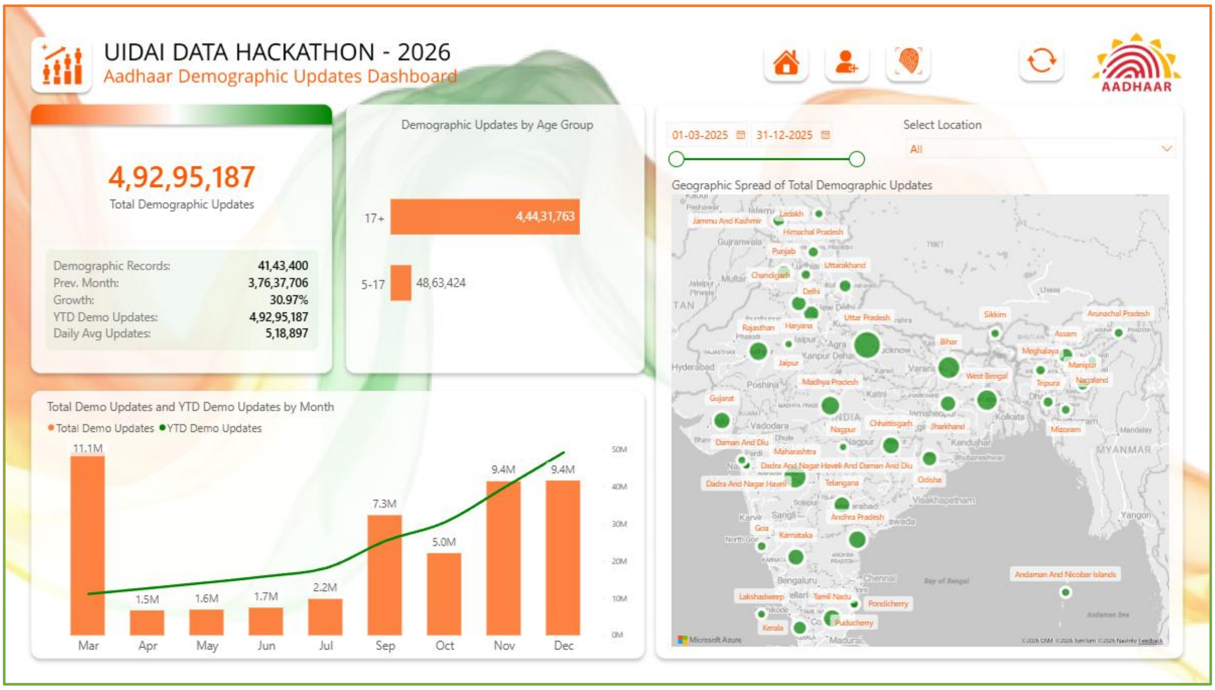
6.1 Aadhaar Enrolment Dashboard Summary



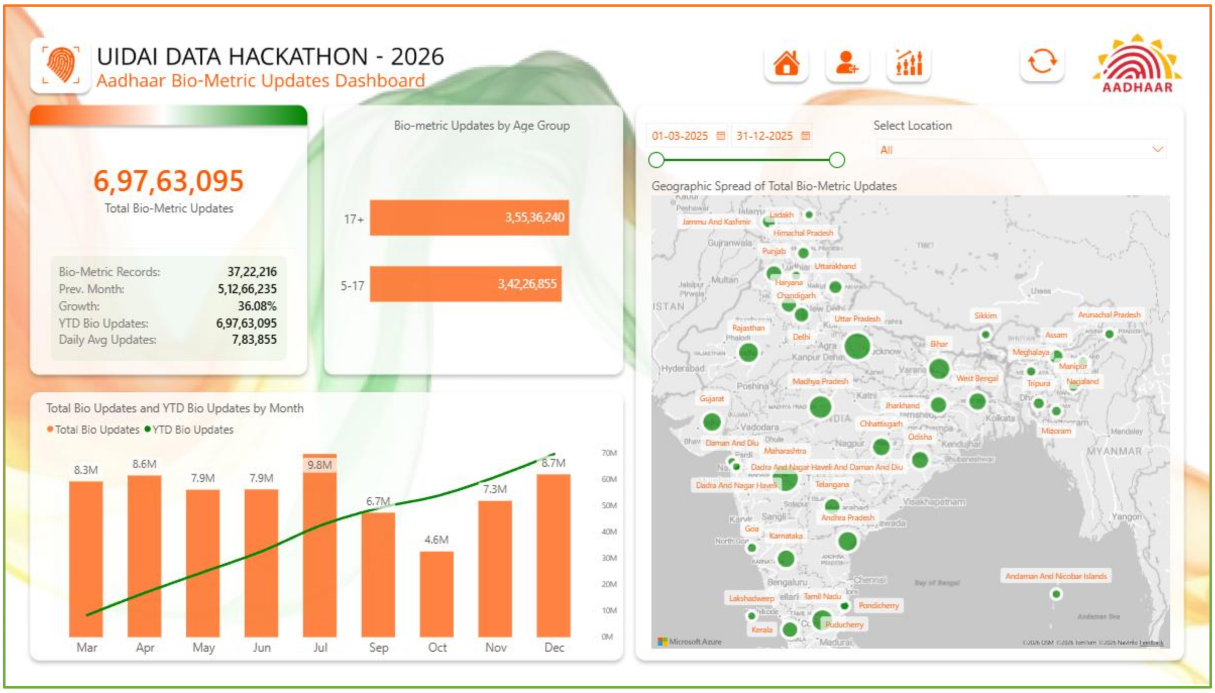
6.2 Aadhaar Enrolment Dashboard



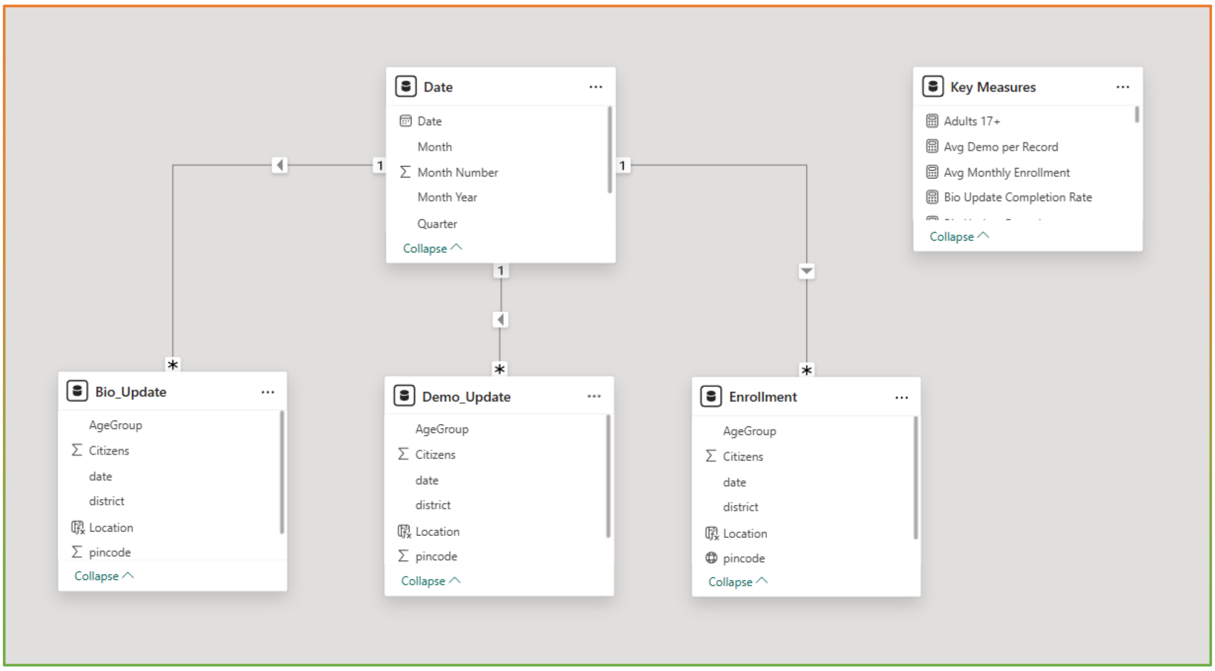
6.3 Aadhaar Demographic Updates Dashboard



6.4 Aadhaar Bio-Metric Updates Dashboard



6.5 Data Model



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